

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 115,000 over 48 months is declined under the default policy (v1.0). Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,143 gives a FOIR of 14.4% (limit 50.0%) and an LTV of 95.8% (limit 80.0%). Risk band: LOW. Reason codes: HIGH_LTV.

For the borrower:

Ravi Kumar, we could not approve the loan of Rs 115,000 as requested because the loan is 95.8% of the vehicle price (our limit is 80.0%). The smallest change that would make it approvable is to reduce loan Rs 115,000 -> Rs 92,000, which brings the instalment to Rs 2,514 and FOIR to 12.8%.

Key drivers:

- FOIR 14.4% vs limit 50.0%
- LTV 95.8% vs limit 80.0%
- Verified income Rs 39,150 (volatility 4.1%)
- HIGH_LTV: LTV 95.83% exceeds policy limit 80.00%

Next steps:

- Offer the borrower the approvable alternative: reduce loan Rs 115,000 -> Rs 92,000.
- Share the plain-language explanation with the borrower.

1. DECISION

Decision: DECLINED Confidence: 0.82

Reason codes: HIGH_LTV

- [blocker] HIGH_LTV: LTV 95.83% exceeds policy limit 80.00%

1a. PATH TO APPROVAL (automatic what-if on the immutable base run)

Minimum change: reduce loan Rs 115,000 -> Rs 92,000

change.loan_amount: 92,000.00

resulting EMI 2,514.04 FOIR 0.1281 LTV 0.7667 -> APPROVED

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is declined because a hard policy rule failed: LTV 95.8% vs limit 80.0% (+15.8 pp over).

The smallest fix is to reduce loan Rs 115,000 -> Rs 92,000, which would bring FOIR to 12.8% and the EMI to Rs 2,514, turning the outcome into approved.

The model's risk band is LOW (score 0.2994 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: ltv (loan / asset value) (95.8% vs a typical 68.1%, +0.050); repayment history score (70.0% vs a typical 71.7%, +0.003).

What pulls risk down: foir (obligations / income) (14.4% vs a typical 41.1%, -0.080); emi-to-income ratio (8.0% vs a typical 34.4%, -0.040); employment stability (95.9% vs a typical 71.8%, -0.024).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 14.4% vs limit 50.0% (-35.6 pp under)

FAIL LTV 95.8% vs limit 80.0% (+15.8 pp over)

PASS age 29 vs minimum 21 (+8 years)

PASS verified monthly income Rs 39,150

risk score 0.299382 = base 0.437336 + sum SHAP -0.137954 (method additive_exact, additivity error 0.0)

phi(foir) = -0.0800 [value 14.4% vs baseline 41.1%]

phi(ltv) = +0.0499 [value 95.8% vs baseline 68.1%]

phi(emi_to_income) = -0.0395 [value 8.0% vs baseline 34.4%]

phi(employment_stability) = -0.0241 [value 95.9% vs baseline 71.8%]

phi(income_volatility) = -0.0183 [value 4.1% vs baseline 22.4%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-LTV-004

full_name: Ravi Kumar

age: 29

location_tier: tier_2
employment_type: gig_platform
declared_monthly_income: 39,000.00
loan_request.amount: 115,000.00
loan_request.down_payment: 5,000.00
loan_request.tenure_months: 48
loan_request.vehicle_price: 120,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Ravi Kumar conf 0.95 [application_form p.None] auto
monthly_income = 39,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 115,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 120,000.00 conf 0.95 [application_form p.None] auto
full_name = Ravi Kumar conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 38,900.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 2,500.00 conf 0.92 [bank_statement.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [bank_statement.txt p.1] auto
vehicle_price = 120,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 115,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 39,150.00
existing_obligations: 2,500.00
emi: 3,142.54
foir: 0.1441
ltv: 0.9583
income_volatility: 0.0408
income_stability: 0.9592
loan_to_income: 0.2448
emi_to_income: 0.0803
input.annual_rate: 14.00
input.existing_monthly_obligations: 2,500.00
input.loan_amount: 115,000.00
input.monthly_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]
input.tenure_months: 48
input.vehicle_price: 120,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.2994 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 14% is comfortably serviceable
risks: LTV 96% implies thin collateral cover
risks: INCOME_MISMATCH: Monthly income differs across documents by 0.3%
risks: LOAN_VS_INVOICE: Loan amount is 95.8% of vehicle invoice value
reasoning: Deterministic FOIR 14.41%, LTV 95.83%; XGBoost band LOW (score 0.2994).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.